Mathematics: analysis and approaches
Standard level
Paper 1
Topic 4 Statistics — box-and-whisker diagrams

8 November 2026

W4ss Academy practice set 3 | **Topic** Topic 4 Statistics — box-and-whisker diagrams | **Ref.** W4SS–MAA–QB–P1–03

Student name

20 minutes

Instructions to students

- Write your name in the boxes above.
- Do not open this examination paper until instructed to do so.
- You are **not** permitted access to any calculator for this paper.
- This paper has a Section A only. Answer all questions. Answers must be written within the answer boxes provided.
- Unless otherwise stated in the question, all numerical answers should be given exactly or correct to three significant figures.
- A clean copy of the **mathematics: analysis and approaches formula booklet** is required for this paper.
- The maximum mark for this examination paper is **[16 marks]**.

Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

Section A

Answer **all** questions. Answers must be written within the answer boxes provided. Working may be continued below the lines, if necessary.

1. [Maximum mark: 4]

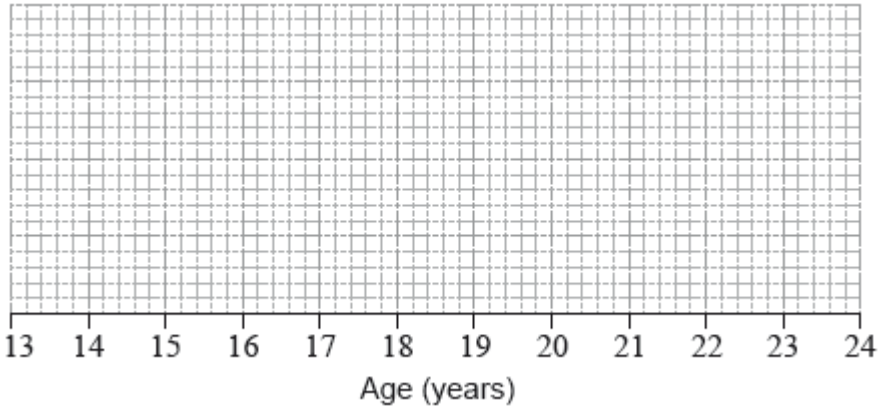
A group of 20 students travelled to a gymnastics tournament together. Their ages, in years, are given in the following table.

Age (years)	14	15	16	17	18	19	20	22
Frequency	1	2	7	1	4	1	1	3

The lower quartile of the ages is 16 and the upper quartile is 18.5.

(a) For the students in this group write down the median age. [1]

(b) Draw a box-and-whisker diagram, for these students' ages, on the following grid.



[3]

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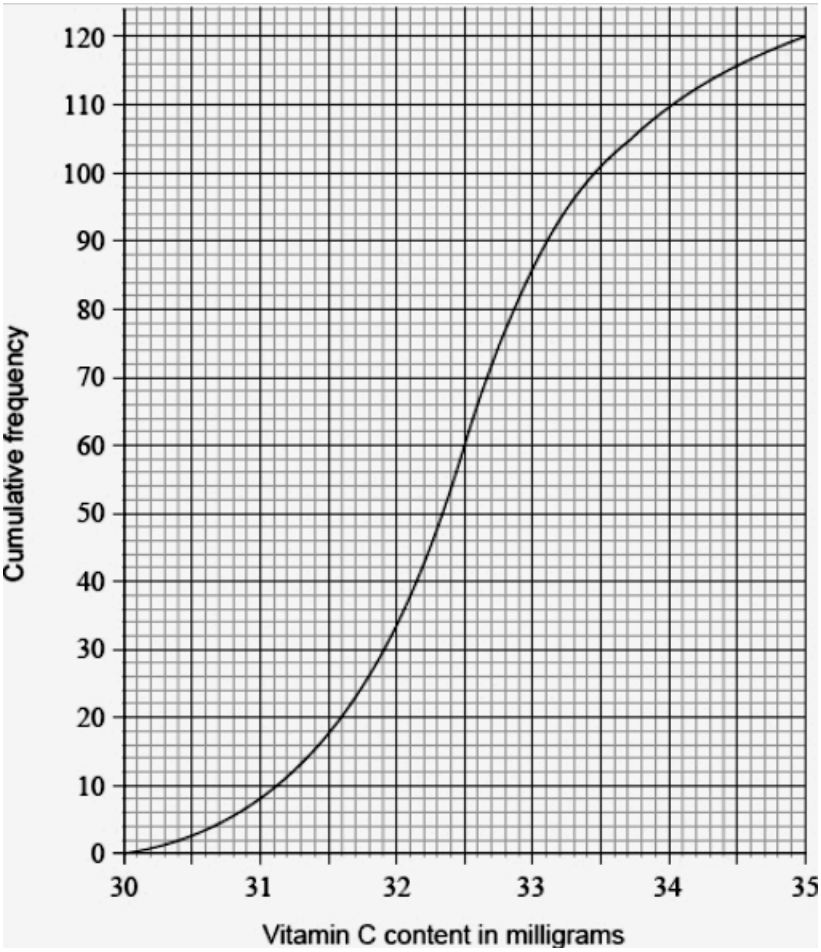
(Question 1 continued)

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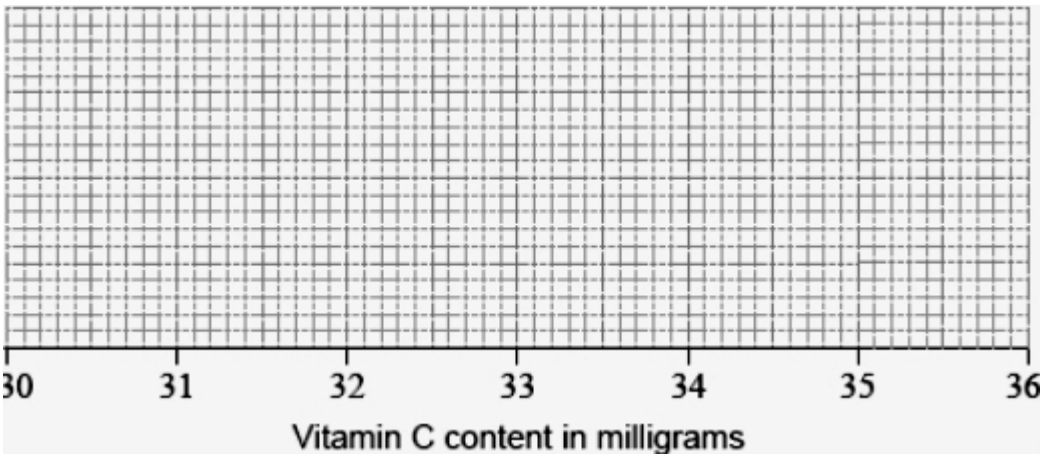
2. [Maximum mark: 3]

A sample of 120 oranges was tested for Vitamin C content. The cumulative frequency curve below represents the Vitamin C content, in milligrams, of these oranges.



The minimum level of Vitamin C content of an orange in the sample was 30.1 milligrams. The maximum level of Vitamin C content of an orange in the sample was 35.0 milligrams.

Draw a box-and-whisker diagram on the grid below to represent the Vitamin C content, in milligrams, for this sample.

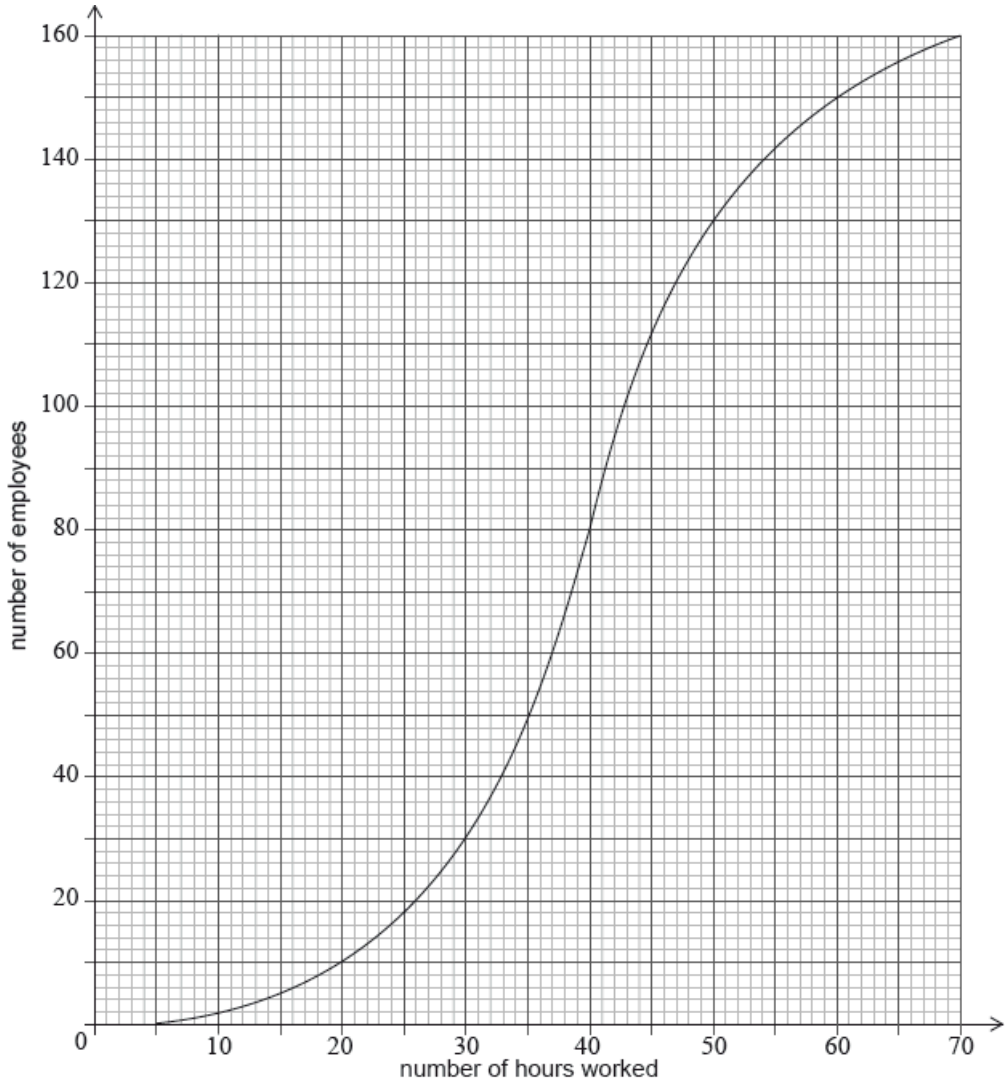


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(Question 2 continued)

3. [Maximum mark: 2]

A city hired 160 employees to work at a festival. The following cumulative frequency curve shows the number of hours employees worked during the festival.



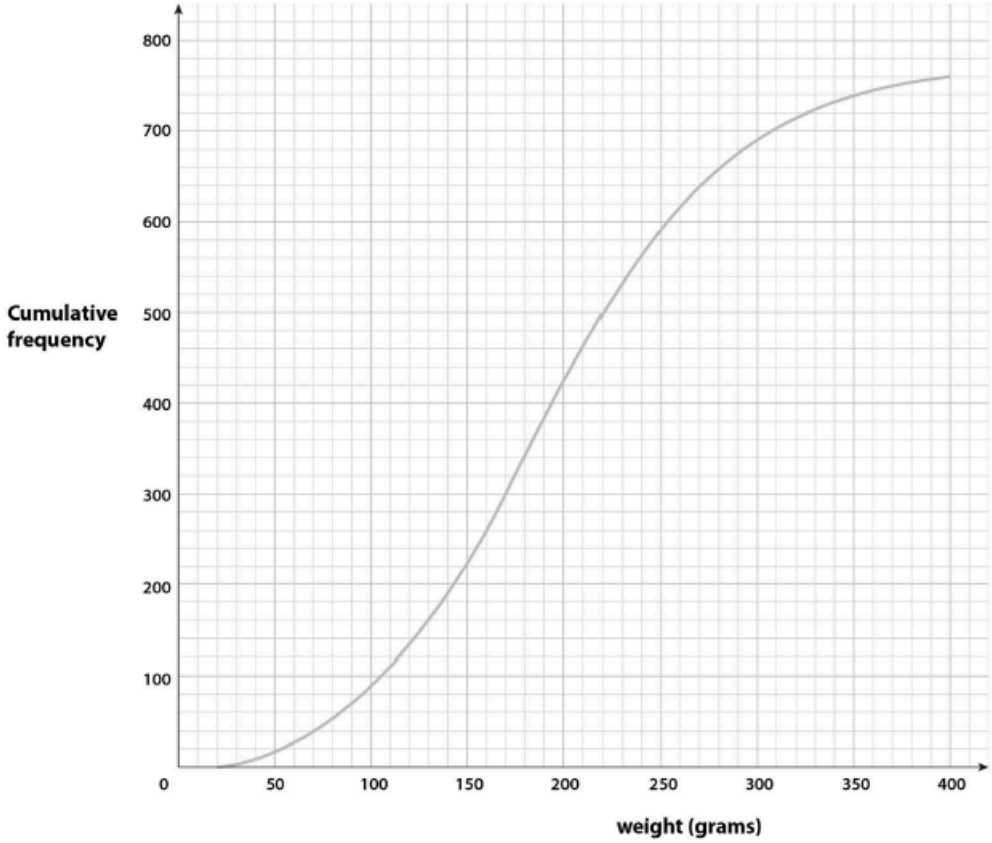
The city paid each of the employees £8 per hour for the first 40 hours worked, and £10 per hour for each hour they worked after the first 40 hours.

(a) Write down the number of employees who worked 50 hours or less. [1]

(b) Find the amount of money an employee earned for working 40 hours. [1]

4. [Maximum mark: 7]

A food scientist measures the weights of 760 potatoes taken from a single field and the distribution of the weights is shown by the cumulative frequency curve below.



(a) Find the number of potatoes in the sample with a weight of more than 200 grams. [2]

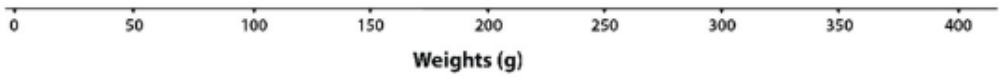
(b) (i) Find the median weight. [1]

(ii) Find the lower quartile. [1]

(iii) Find the upper quartile. [1]

The weight of the smallest potato in the sample is 20 grams and the weight of the largest is 400 grams.

(c) Use the scale shown below to draw a box and whisker diagram showing the distribution of the weights of the potatoes. You may assume there are no outliers.



[2]

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(Question 4 continued)

Sources of the questions

This practice paper was put together by W4ss Academy for internal class use. The questions are taken from the IB DP Mathematics: analysis and approaches Questionbank (standard level, topic 4 Statistics and probability), as chosen in the class QuestionBank test 3. Some of them come from former courses (Mathematics SL, Mathematical studies SL). They remain the property of the International Baccalaureate Organization and must not be published or sold.

Section A — question 1	17N.1.SL.TZ0.T_1 (former Mathematical studies SL course) · 4.2, 4.3 median, box-and-whisker diagram — part (a)(ii) of the original, relabelled (a): part (a)(i) is not in the test
Section A — question 2	16N.1.SL.TZ0.T_2 (former Mathematical studies SL course) · 4.2 cumulative frequency curve, box-and-whisker diagram — the part of the original that is in the test (figures taken from screenshots)
Section A — question 3	17M.1.SL.TZ2.S_8 (former Mathematics SL course) · 4.2 cumulative frequency curve — parts (a)(ii) and (b)(i) of the original, relabelled (a) and (b)
Section A — question 4	EXN.1.SL.TZ0.4 · 4.2 cumulative frequency curve, quartiles, box-and-whisker diagram

Please **do not** write on this page.

Answers written on this page
will not be marked.